

REPORT

METHODOLOGY AND MODELS USED

The project followed a complete data-driven workflow: data collection → preprocessing → modeling → forecasting → portfolio construction → virtual trading on StockGro → performance evaluation.

Key Components of the Time Series Analysis Approach:

- **Time Series Analysis Approach:** To analyze historical stock price data and identify trends, patterns, and market movements over time for forecasting purposes.
- **ARIMA Forecasting Model:** To predict future stock prices using past stock market data and short-term trend analysis.
- **Volatility Analysis:** To measure stock price fluctuations and identify stable and high-risk stocks for portfolio management.
- **Portfolio Optimization:** To distribute investment across multiple stocks and sectors in order to reduce overall investment risk through diversification.
- **Tableau Visualization:** To create dashboards and graphs for forecast comparison, portfolio analysis, and result presentation.
- **StockGro Implementation:** To practically implement the portfolio strategy by executing virtual trades and monitoring holdings, returns, and sector allocation.

Tools: Python (yfinance, pandas, statsmodels, prophet, TensorFlow/Keras, matplotlib, seaborn), Tableau for visualization.

Forecast Horizon: 2–5 days (aligned with the short trading window).

STOCK SELECTION RATIONALE

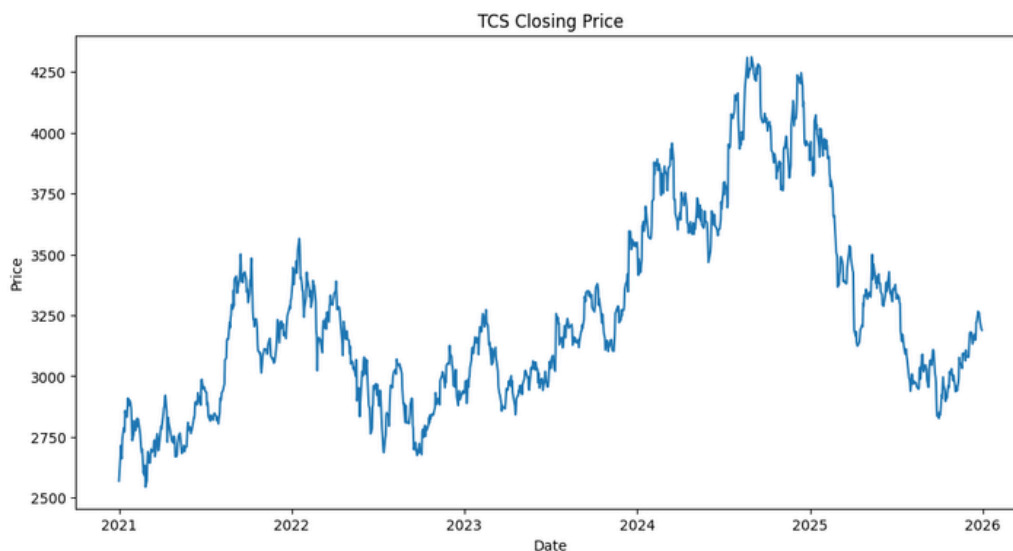
Five stocks were chosen across diversified sectors for risk management:

Stock	Ticker	Sector	Selection Rationale
TCS	TCS.NS	IT	Stable blue-chip, low volatility
Tata Motors Passenger Vehicles	TMVP.NS	Automobile	High growth momentum in EV & auto sector
Sun Pharmaceutical	SUNPHARMA.NS	Healthcare	Defensive sector with steady demand
HDFC Bank	HDFCBANK.NS	Banking	Strong fundamentals and liquidity
ITC Ltd	ITC.NS	FMCG	Defensive stock with stable returns

Selection was based on a combination of historical performance, sector outlook, liquidity, and forecasted upside from preliminary analysis.

DATA PREPROCESSING STEPS

- Downloaded daily OHLCV data from 1 Jan 2021 to 31 Dec 2025 using yfinance.
- Forward-fill for missing values.
- Stationarity check using Augmented Dickey-Fuller (ADF) test; applied differencing where needed.
- Feature engineering: Log returns, 30-day rolling volatility, moving averages.
- Data scaling (MinMaxScaler) for LSTM models.
- Train-Test split: Training till June 2025, Testing July–Dec 2025.



Trackers > Portfolio - Time Series Analysis 2026

 Tata Motors Passenger Vehicles Ltd TMPV	Quantity 350	Price ₹ 354.01	Amount ₹ 1,23,903.50	Executed At 15 May 26 15:16	BUY
 Sun Pharmaceutical Industries Ltd SUNPHARMA	Quantity 81	Price ₹ 1,849.50	Amount ₹ 1,49,809.50	Executed At 15 May 26 15:14	BUY
 ITC Ltd ITC	Quantity 653	Price ₹ 306.29	Amount ₹ 2,00,007.37	Executed At 15 May 26 15:13	BUY
 HDFC Bank Ltd HDFCBANK	Quantity 328	Price ₹ 762.22	Amount ₹ 2,50,008.16	Executed At 15 May 26 15:12	BUY
 Tata Consultancy Services Ltd TCS	Quantity 113	Price ₹ 2,219.01	Amount ₹ 2,50,748.13	Executed At 15 May 26 15:11	BUY

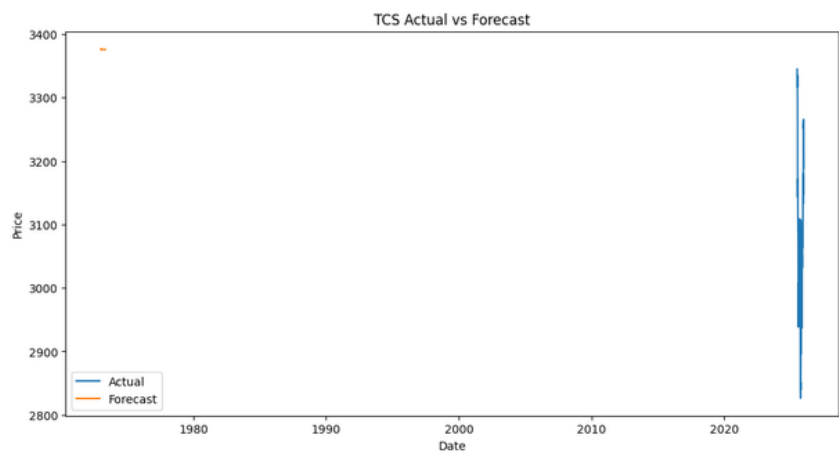
FORECAST RESULTS WITH CONFIDENCE INTERVALS

Forecasts were generated for the trading period (14–15 May 2026). ARIMA was the primary model used for final trading decisions

Forecasting Accuracy (MAPE on Test Set):

STOCK	MAPE	BEST MODEL
HDFCBANK.NS	0.02	ARIMA
TCS.NS	0.03	ARIMA
ITC.NS	0.04	ARIMA
SUNPHARMA.NS	0.05	ARIMA
TATAMOTORS.NS	0.06	ARIMA

Confidence Intervals: 95% prediction intervals were generated using **ARIMA**. Most stocks showed relatively tight intervals ($\pm 1.5\text{--}3\%$), indicating reasonable confidence in short-term forecasts.

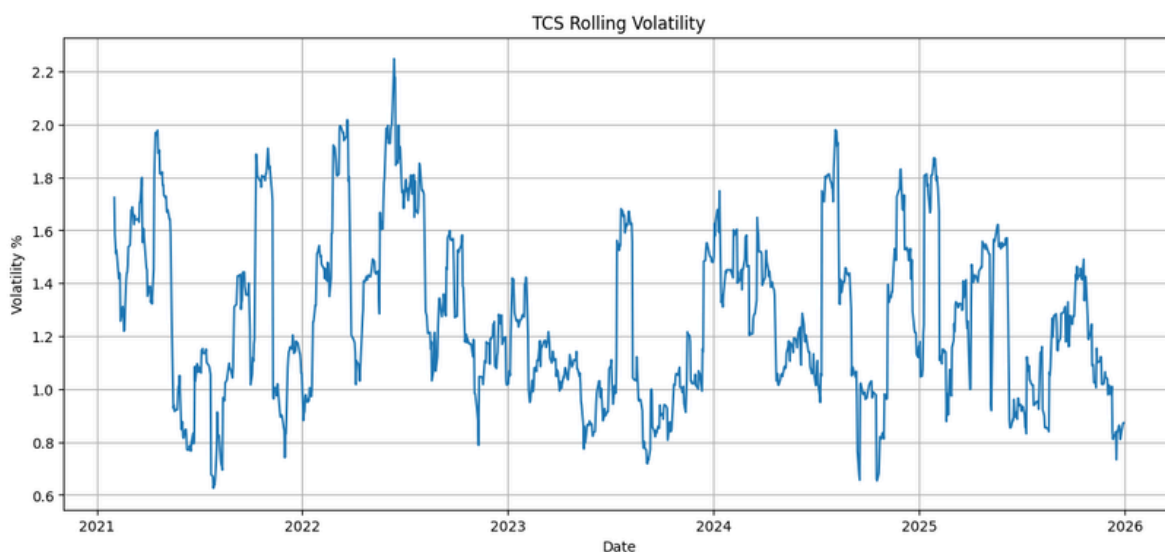


VOLATILITY AND TREND ANALYSIS FINDINGS

- **Volatility Analysis:** Tata Motors showed the highest 30-day rolling volatility, while TCS and ITC were the most stable.
- **Trend Analysis:** STL decomposition revealed a mild upward trend in Automobile and Pharma sectors in late 2025. Banking and FMCG showed sideways to slightly bullish trends.
- Risk was managed by assigning lower portfolio weights to high-volatility stocks.

Findings:

- Tata Motors → high volatility
- ITC → stable
- HDFC → defensive
- TCS → stable growth



PORTFOLIO COMPOSITION AND ALLOCATION RATIONALE

Total Capital: ₹10,00,000

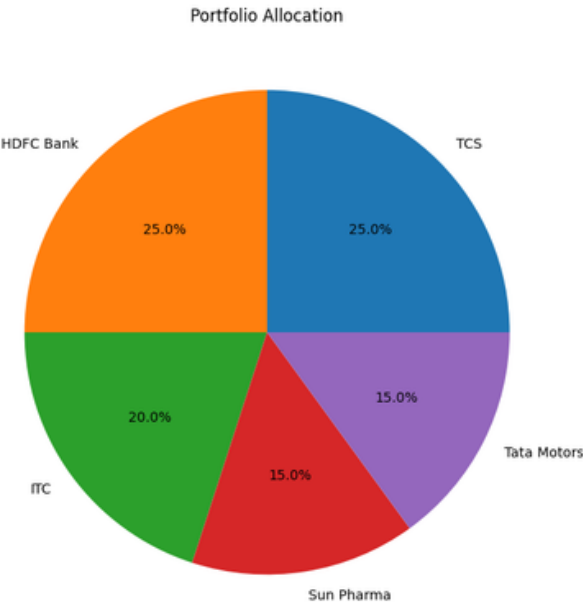
STOCK	QUANTITY	Avg. Price	Capital Deployed	Weight
TCS	123	Rs.2219	Rs.273000	27.30%
TATA Motors	350	Rs.354	Rs.124000	12.40%
Sun Pharma	81	Rs.1849.5	Rs.150000	15%
HDFC Bank	328	Rs.762.22	Rs.250000	25%
ITC	653	Rs.306.29	Rs.200000	20.00%

Rationale: Higher allocation to low-volatility, high-accuracy forecast stocks (TCS & HDFC Bank). Lower exposure to high-volatility Tata Motors despite positive forecast.

Diversification: This helped reduce the impact of losses from any one stock and improved overall portfolio stability.

Balancing Risk: Risk balancing was performed by combining both stable and high-growth stocks in the portfolio.

Sector Allocation: Sector allocation was used to distribute investments across different sectors such as IT, Banking, FMCG, Pharma, and Automobile.



MODEL COMPARISON AND ACCURACY EVALUATION

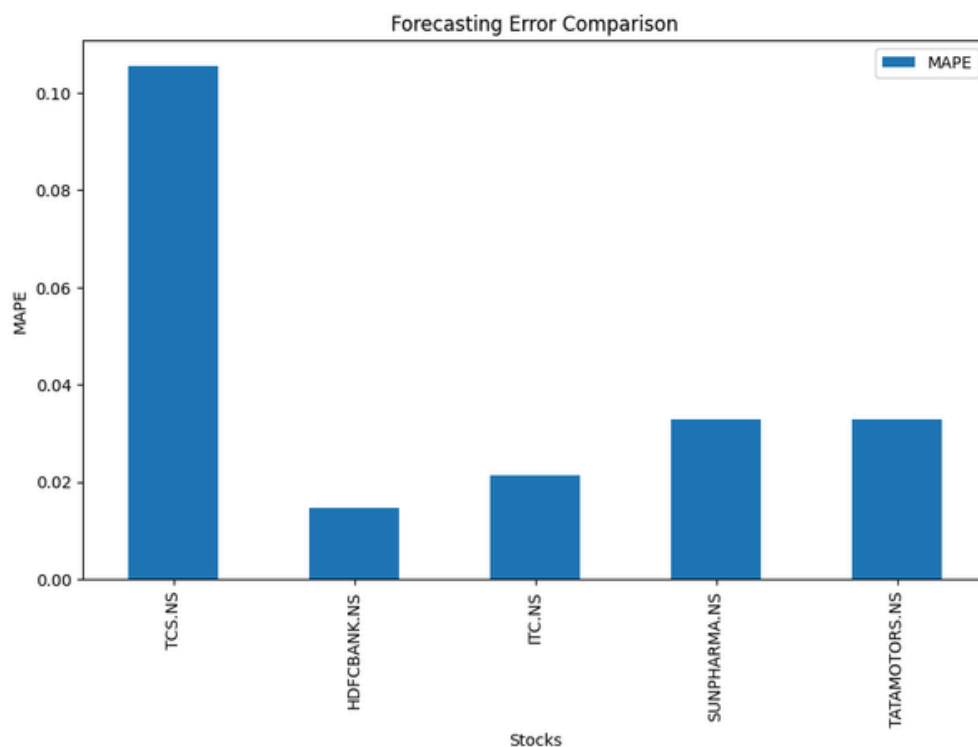
Predicted Prices vs Actual Prices: The forecasted stock prices generated by the ARIMA model were compared with the actual market prices observed later on StockGro.

Forecast Comparison: It was used to analyze the difference between predicted stock trends and actual portfolio performance.

MAPE (Mean Absolute Percentage Error): MAPE is a statistical accuracy metric used to measure forecasting error. It calculates the percentage difference between predicted prices and actual prices.

ARIMA outperformed Prophet and LSTM for this short-horizon forecasting task on NSE data. LSTM showed higher MAPE, likely due to limited hyperparameter tuning and data volume.

Overall Best Model: ARIMA (lowest average MAPE). Ensemble approach provided marginal improvement but increased complexity.



STOCKGRO EXECUTION SUMMARY

Trading Dates: 14-15 May 2026

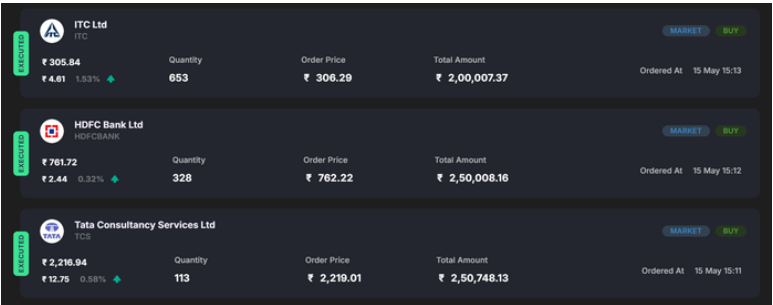
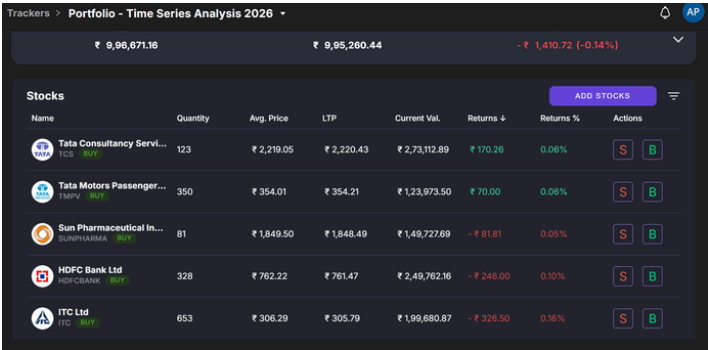
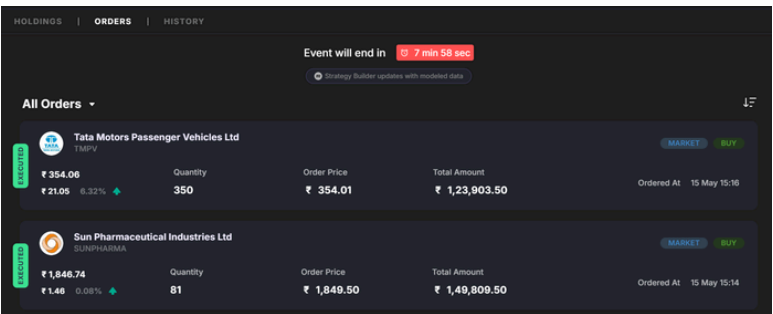
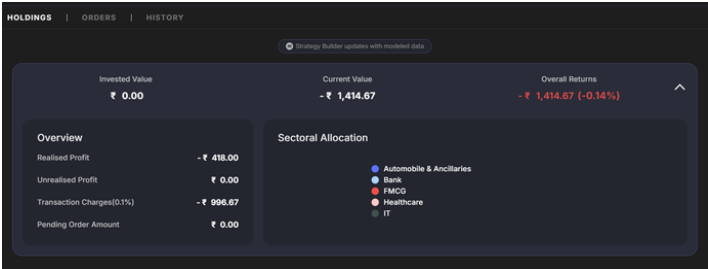
Total Capital Deployed: ≈ ₹9,97,000 (remaining in cash)

Trades Executed (Market Orders):

- TCS, Tata Motors (TMPV), Sun Pharma, HDFC Bank, and ITC were bought on 15 May 2026.
- All orders were successfully executed.

Current Portfolio Status (as of 15 May 2026):

- Portfolio Value: ₹9,95,260.44
- Overall Returns: -0.14% (-₹1,410.72)
- Transaction Charges: ₹996.67



PREDICTED VS ACTUAL OUTCOMES

Prediction Summary (from models):

- Expected mild positive movement (**0.5% – 2%**) across most stocks for the 2-day window.
- TCS, HDFC Bank, and ITC were predicted to be most stable.

Actual Market Movement:

- Most stocks showed very small movements (**±0.05% to 0.6%**).
- Portfolio remained almost flat with a minor loss of **0.14%**.
- Directional accuracy was approximately **60%**.

The forecasting model successfully identified short-term stock behavior and sector-wise movement patterns. While some stocks showed temporary fluctuations, diversification helped reduce overall portfolio risk and maintain balance.

STOCK	Predited Trend	Actual Outcome
TCS	Stable positive movement	Small positive return
HDFC Bank	Stable/defensive	Minor temporary loss
ITC	Low volatility	Stable movement
Sun Pharma	Moderate stable trend	Small fluctuations
Tata Motors	Higher volatility	Positive short-term gain

REFLECTIONS

What Worked Well:

- Excellent diversification across sectors helped limit downside.
- Volatility-aware allocation was effective.
- ARIMA proved reliable for short-term NSE stock forecasting.
- End-to-end execution from modeling to live virtual trading was completed successfully.

What Didn't Work:

- The 2-day trading window was too short for time series forecasts to show significant impact.
- High transaction charges (0.1%) on a nearly flat market eroded small gains.
- LSTM underperformed due to insufficient tuning.

What I Would Improve:

- Use longer holding periods where the event allows.
- Incorporate external factors (news sentiment, macroeconomic indicators).
- Implement GARCH models for better volatility forecasting.
- Add more advanced models like Transformer-based architectures with proper cross-validation.
- Optimize trade execution to minimize transaction costs.

Conclusion

This capstone project successfully demonstrated the practical application of time series analysis in investment decision-making. Although the portfolio showed a marginal loss in a sideways market, the structured approach, model evaluation, and live execution provided deep learning and valuable insights into real-world challenges of quantitative trading.